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Differences in the neighborhood: Structural variations in the carapace of shore crabs *Carcinus maenas* (Decapoda: Portunidae)

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ABSTRACT

The present study compared the carapace structure of *Carcinus maenas* in two nearby sites (2 km apart) within Minho estuary, submitted to different physicochemical and ecological conditions (water temperature, pH, crabs' density and sex ratio). The carapace structure of the carapace and chelae of the crabs presented significant differences between sampling sites (*t*-test; $p < 0.01$). The SIMPER analysis revealed that the Weight/CW and Thickness/CW ratios explained all the dissimilarities found among sites. Overall, the male carapace was proportionally thicker at station 2 (*t*-test; $p < 0.01$), while the female carapace was proportionally thicker at station 1 (*t*-test; $p < 0.001$). A thicker carapace can be advantageous when competing for food or a sexual partner. We hypothetized that, since at station 2, the density of individuals was twice higher than at station 1, it is likely that agonistic encounters are more frequent, thus favouring a thicker carapace.

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1. Introduction

The crustacean hard, calcified exoskeleton has multiple functions: body support, resistance to mechanical loads, environmental protection and resistance to desiccation (Vincent, 1991, 2002; Vincent and Wegst, 2004; Sanchez et al., 2005). However, the rigidity of the carapace restricts growth, once any increase in size requires the shedding of the exoskeleton. Therefore, crustaceans present a discontinuous growth pattern, characterized by successive ecdysis (periodic shedding of the exoskeleton) over time (Hartnoll, 2001).

Moulting is a complex process, affected by a range of environmental cues and regulated by a cascade of hormonal signals involving changes in gene expression, cellular commitment, mitotic and secretory activity, endocrinology, behavior and cell death (Loeb, 1993). Several environmental and ecological factors may be involved in the moulting process, such as water temperature, salinity, photoperiod, tidal cycle, food availability and population clusters (Chang, 1995; Hartnoll, 2001). Also, several hormones are involved in the moulting process control (Lachaise et al., 1993; Hartnoll, 2001), thus moulting results from complex interactions between endogenous regulation and environmental inputs and involves hormonal, neurohormonal and enzymatic control (Styrishave et al., 2004).

The green or shore crab *Carcinus maenas* is a common inhabitant of intertidal systems worldwide. It is indigenous to European and Northern Africa coasts and estuaries, spanning *ca.* 49° of latitude (from Norway to Mauritania) (Roman and Palumbi, 2004). Recently, this species has also colonized several systems in Argentina, Australia, Japan, South Africa and both coasts of North America (Brenchley, 1982; Lafferty and Kuris, 1996; Behrens Yamada, 2001; Hidalgo et al., 2005) and has, therefore, been considered one of the worst invasive species in the world (Lowe et al., 2000).

During its life cycle, *Carcinus maenas* performs about 18 moultings, four of them on the first developmental stages (zoea and megalopa) and fourteen after larval stage. Shore crabs grow up and moult until the terminal ecdysis (cessation of growth), which occurs when they reach 86 mm of carapace width (CW) (Crothers, 1967). Moreover, the shore crab takes about 10 moults to reach 20 mm carapace width (CW) in its first year, if conditions are favorable (Crothers, 1967). *C. maenas* may moult more than once per year after the first year if conditions are good but moulting rate slows once maturity is reached, and is probably about once per year post maturity (Crothers, 1967).

Considerable variability in morphology, physiology and behavior is commonly found in *Carcinus maenas* in relation to sex, size and carapace coloration (e.g. McGaw and Naylor, 1992; Reid

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et al., 1994; Ledesma et al., 2010). This intraspecific variability reflects the phenotypic adaptive responses of individual crabs in relation to their ability to withstand environmental variability (McGaw and Naylor, 1992; Warman et al., 1993; Abelló et al., 1997). Differences in morphology, physiology and/or behavior are often displayed when organisms are submitted to different environmental and ecological conditions (e.g. Reid et al., 1997). However, the mechanisms that lead to these differences are seldom investigated in the field. While the effect of environmental variables can be easily replicated in the laboratory, ecological processes are hard to reproduce. In this context, this study aimed to compare the carapace structure of the shore crab sexes (males and females) and color morphotypes (green and red) among different sampling sites. The comparisons were made in two nearby areas within the same ecosystem, and with distinct habitat utilization by crabs, reflected in differential densities, sex ratio and environmental conditions (ATS personal observation).

2. Materials and methods

2.1. Sampling and laboratory analyses

This study was conducted in the Minho Estuary (NW of the Iberian Peninsula), which ranges up to 40 km (at the upstream limit of spring tides), covering a total area of 23 km^2. The Minho estuary is a very shallow system (Moreno et al., 2005), with a mean depth of 2.6 m (Freitas et al., 2009) and characterized as a mesotidal and partially mixed system, although during periods of high river flow, it tends towards a salt wedge estuary (Sousa et al., 2005). The present study was carried out in the lower estuary (the first 8 km from the estuary mouth).

Carcinus maenas individuals were collected in April 2010 in two nearby sites (approximately 2 km apart) in the lower Minho estuary (Fig. 1). The two sites were chosen in order to assess possible dissimilarities on carapace structure related to *Carcinus maenas* density, sex ratio and environmental variables. Both sampling stations are located in the subtidal zone. However, during low water, at station 2 the water column is very shallow (*ca.* 20 cm), while at station 1 water level reaches *ca.* 1 m of minimum depth. Moreover, station 1 is closer to the river mouth, with the soft bottom often sparsely covered by debris, such as drifting seaweeds and empty mollusc shells, while station 2 is located inside the Coura salt marsh, dominated by the seagrass *Spartina maritima* and its soft bottom houses an especially abundant but poorly diverse fauna represented mainly by crabs *C. maenas*, gobies *Pomatoschistus* spp., common shrimps *Crangon crangon* and the peppery furrow shells *Scrobicularia plana* (ATS personal observation). At each site, three samples were taken using a 1 m beam trawl (5 mm mesh size) towed at constant speed (2 km h^{-1}) for 2–3 min, during high water. Additionally, at each site, environmental variables (water temperature, salinity, conductivity, pH and oxidation reduction potential (ORP)) were taken through a multiparameter probe YSI 6820 on the surface (two replicates) and 20 cm above the bottom (two replicates). For statistical analysis, four measurements of each site (2 × surface and 2 × bottom) were used to assess differences between sampling stations.

All collected *Carcinus maenas* individuals were counted in order to assess its density on the two sampling sites. Intermoult crabs were sexed (males or females) according to their abdominal anatomy (see Crothers, 1967) and classified into color morphotypes (green or red), based on the predominant color of the thoracic sternum (McGaw and Naylor, 1992). The carapace width (CW) and propodus length (PL) of right and left claw were then measured to

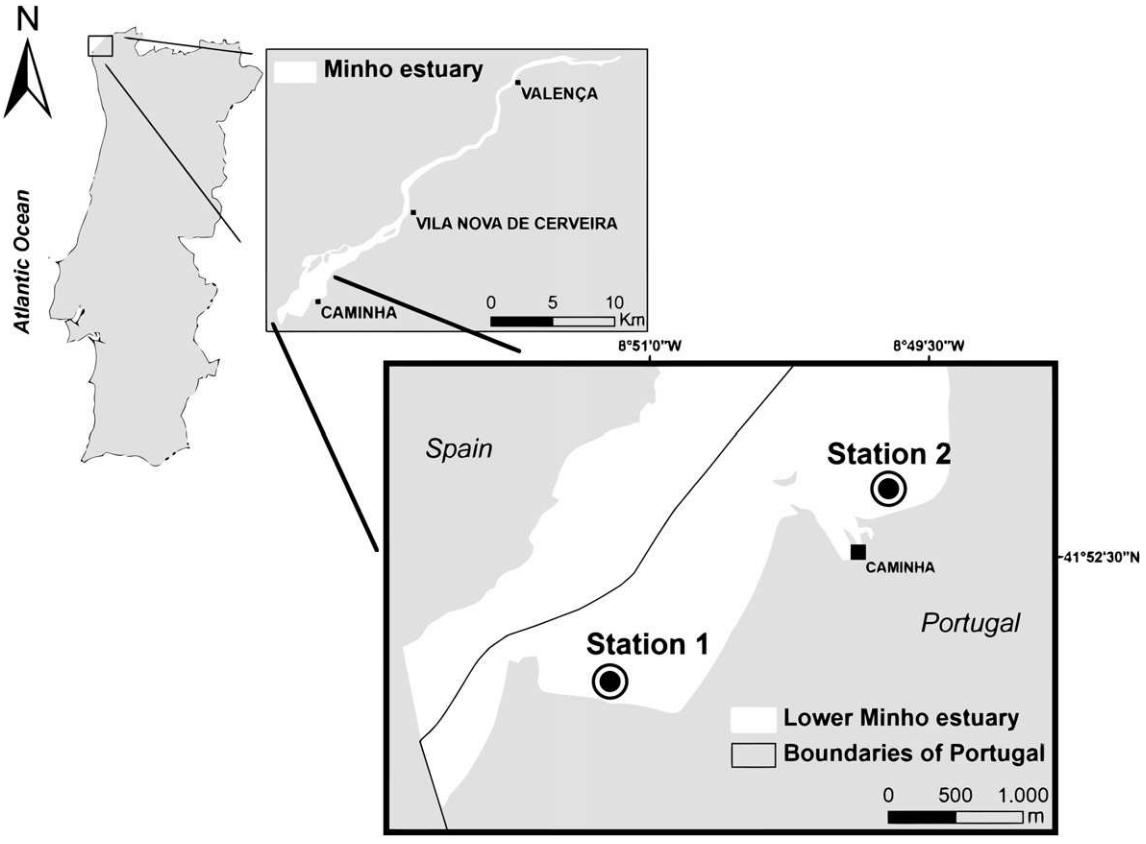


Fig. 1. Map of the study site showing the two sampling stations on the lower Minho Estuary.

the nearest 0.01 mm using a digital caliper. Carapaces were then air-dried for 4 days and afterwards weighed to the nearest 0.0001 g using a digital scale. A small segment of the carapace (approximately 1 cm^2) was removed from each tested crab, along the fracture line, and its thickness was measured to the nearest 0.01 mm with a digital caliper. A similar procedure was used to assess the claw thickness, which was also air-dried for 4 days, and had a small segment (approximately 0.5 cm^2) removed from the area corresponding to the maximum propodus height on the frontal part of the chelae (Fig. 2). For thickness samples, the average of three separate measurements was recorded, and its mean value used in statistical analysis.

For the comparisons in the carapace structure between mature versus immature crabs, we considered *Carcinus maenas* individuals as sexually mature when they were larger than approximately 30 mm of CW (d'Udekem d'Acoz, 1993; Behrens Yamada et al., 2005).

For all variables, ratios were calculated in order to avoid bias due to increasing carapace thickness with individual growth (i.e. thickness/CW, for this ratio was assumed that the higher the ratio, thicker is the individual; weight/CW, for this ratio was assumed that the higher the ratio, heavier is the individual). The same procedure was used with right (RC) and left chelae (LC) (i.e. right and left propodus length) and chelae thickness (i.e. RC thickness/RC, for this ratio was assumed that the higher the ratio, thicker is the RC of the individual; LC thickness/LC, for this ratio was assumed that the higher the ratio, the thicker is the LC of the individual). Since most of the ratio values presented two or more non-significant decimal digits, a simple multiplication (times 1000) was applied when needed, in order to highlight the numerical differences present. Nevertheless, this process did not affect the calculation, since the same procedure was applied in all cases.

2.2. Data analysis

Data were analyzed with the PRIMER v6® package (Clarke and Warwick, 2001). A one way ANOSIM (analysis of similarity) and an nMDS (non-metric multi-dimensional scaling) were used to independently infer the putative differences among sampling stations, regarding environmental descriptors and *Carcinus maenas* carapace (proportions of CW with thickness and weight, as well as the proportion of chelae thickness with propodus length). All data were normalized, without prior data transformation, in order to avoid bias due to differences in variability range of each variable (Clarke and Warwick, 2001). Due to the character of the data (morphometrics), the similarity matrices were based upon Euclidian distances as recommended by Clarke and Warwick (2001). In case of significance, we used the SIMPER routine to determine the percent contribution of each variable to the observed dissimilarities between sampling stations. Additionally, to examine differences between sampling stations regarding environmental variables and carapace variables between, sex and color morphotype, pairwise comparisons were performed using Student's *t*-test (unpaired). Also, the Pearson's correlation test was used to determine whether a correlation existed between CW with carapace thickness and weight; these analyses were carried out with Statistica 5 software.

3. Results

3.1. Ecological data

Overall, the environmental descriptors were similar in the two sampling stations (ANOSIM; $R = 0.26$; S.L. > 0.05). However, when compared individually, there were significant differences in pH (*t*-test, $p < 0.05$) and temperature of the water (*t*-test, $p < 0.05$), with both variables presenting higher values at station 2. The SIMPER procedure also suggested that those variables were the most important, accounting for near the half of the dissimilarities observed among sampling sites (44.90%).

Crab density at station 1 was 33.5 4.95 ind m^{-2} (mean SD), being 58.2% females, 29.9% males and 11.9% juveniles (CW < 10 mm), while at station 2, density nearly doubled (72.0 6.36 ind m^{-2}), with a predominance of males (56.3%) over females (43.8%), while juveniles were absent. On an annual basis, station 1 is dominated by juveniles and females, whereas at station 2, males are dominant (ATS personal observation).

3.2. Carapace structure: range of observations

A total of 99 individuals (green morphotype = 68.7%; red morphotype = 31.3%) were analyzed, being 43 males (green morphotype = 81.4%; red morphotype = 18.6%) and 56 females (green morphotype = 58.9%; red morphotype = 41.1%). The range of

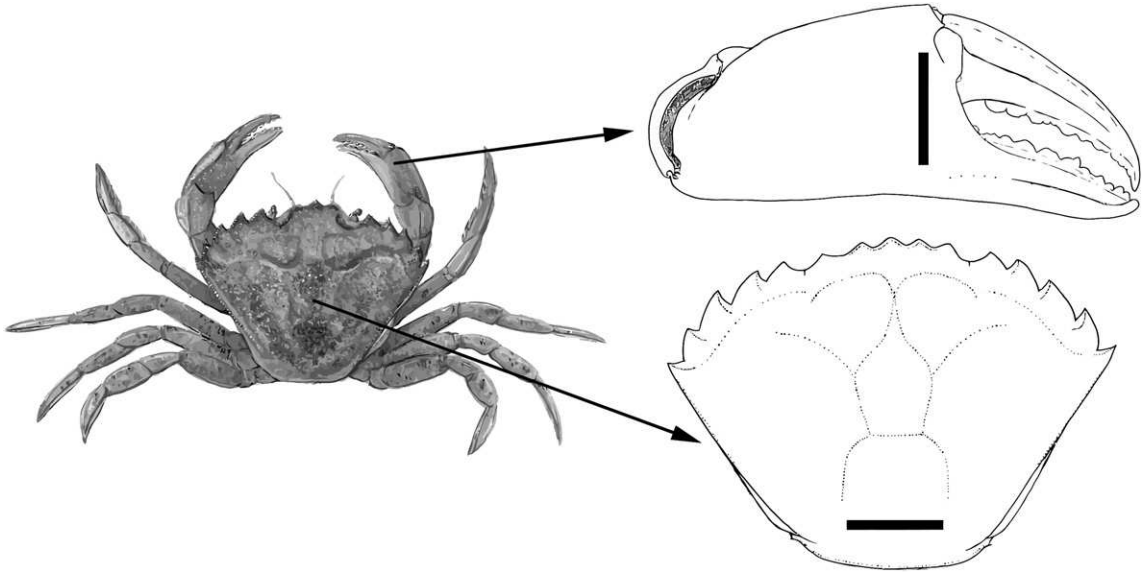


Fig. 2. *Carcinus maenas* areas measured for carapace and claw thickness (black bars).

CW size varied between 14.22 and 65.19 mm, with an average value of 35.58 mm. The carapace thickness varied from 0.07 to 0.39 mm, with an average value of 0.22 mm. Although, in general, individuals showed the right chelae larger than the left one (81.8% of the individuals), this difference was not significant (paired *t*-test; $p > 0.05$).

Significant differences were detected between males and females (both color morphotypes) in four carapace variables, namely, carapace thickness, carapace thickness/CW, RC thickness/RC, LC thickness/LC, indicating that females were proportionally thicker than males (Table 1).

Significant differences between crabs with the red and green morphotypes were detected when regarding the thickness/CW and weigth/CW ratios, with red crabs being proportionally thicker (*t*-test; $p > 0.001$) and heavier (*t*-test; $p > 0.001$) than individuals of the green morphotype.

The Pearson's correlation between carapace thickness and CW was significant either for males ($r^2 = 0.55$; $p < 0.001$) and females ($r^2 = 0.86$; $p < 0.001$). A similar result was obtained when regarding weight and CW, with the Pearson's correlation being significant for males ($r^2 = 0.90$; $p < 0.001$) and females ($r^2 = 0.92$; $p < 0.001$), with the carapace of females being heavier than that for males.

3.3. Carapace structure: mature versus immature crabs

Significant differences between males (*t*-test; $p < 0.01$) and females (*t*-test; $p < 0.001$) were detected on the thickness/CW ratio when divided into mature or immature individuals, with immature males presenting a relatively thicker carapace than immature females, whereas when mature this patterns shifts, with females becoming proportionally thicker than males (Fig. 3). Mature males also presented a larger ratio of chelae size (right and left) with CW than immature males, whilst females did not differ. Regarding the ratio of chelae size with its thickness, males only differed in the left chelae, which was thicker when immature. In contrast, females presented thicker chelae (right and left) when mature. Table 2 sums up the differences detected in the chelae, as well as the cases when the differences were significant.

3.4. Carapace structure: differences between sites

Carcinus maenas individuals of both sexes and color morphotypes differed significantly between sampling sites with respect to the ratios of the carapace width with thickness and weight, as well as the ratios of chelae thickness with propodus length (ANOSIM; $R = 0.49$; S.L. < 0.01). The nMDS ordination also suggested a strong dissimilarity between individuals of different sampling sites (2D Stress = 0.01) (Fig. 4). The SIMPER analysis further confirmed a high dissimilarity between sampling sites which was 100% explained by two variables: ratio thickness/CW (53.86%) and ratio weight/CW (46.14%).

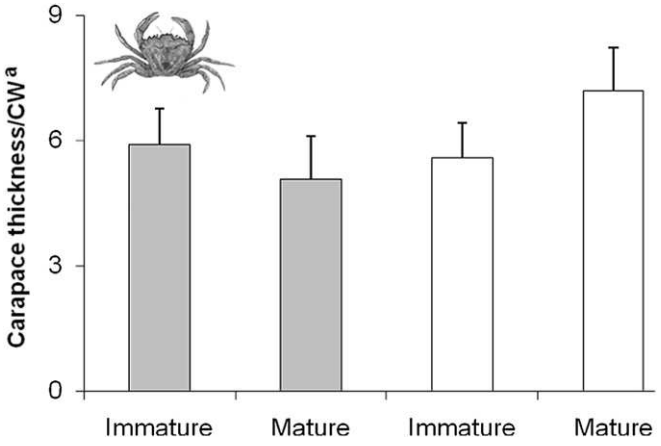


Fig. 3. Differences in the ratio of carapace width (CW) and thickness (mean SD) of *Carcinus maenas* immature (CW < 30 mm) and mature (CW > 30 mm) individuals in Minho estuary. Grey bars, males; white bars, females; [a], ratio times 1000.

Male and female crabs significantly differed between sites in respect to the ratio of carapace weight and width (*t*-test; $p < 0.001$). Moreover, the ratio of carapace thickness and width of males (*t*-test; $p < 0.01$) and females (*t*-test; $p < 0.001$) differed significantly between sampling sites (Fig. 5).

Table 3 shows the differences and its significance levels between sampling sites for both males and females crabs. Mature males of station 1 presented a smaller ratio (i.e. proportionally larger chelae) of chelae size (right and left) with CW than males from station 2, whilst females did not differ. Regarding the ratio of chelae size with its thickness, males only differed in the left chelae, presenting thicker left chelae at station 2. Conversely, females presented thicker chelae both right and left at station 1.

4. Discussion

There was a relatively large variability in the carapace structure of *Carcinus maenas* between functional groups i.e. sexes, color morphotypes and life stages, as well as among sampling stations.

4.1. Differences among Carcinus maenas functional groups

Carcinus maenas males and females presented a notably different pattern in their carapace structure, both in their

Table 1
Carapace variables and ratios (mean SD) of *Carcinus maenas* color morphotypes and sexes in Minho estuary. S.L, significance level; CW, carapace width; PL, propodus length; RC, right chelae; LC, left chelae. NS, non-significant; *, S.L. < 0.05; **, S.L. < 0.01; ***, S.L. < 0.001.

Variable	Green			Red		
	Male	Female	S.L.	Male	Female	S.L.
CW (mm)	38.62 ± 10.55	34.83 ± 10.03	NS	37.59 ± 10.20	34.13 ± 10.26	NS
Carapace weight (g)	0.79 ± 0.47	0.63 ± 0.46	NS	0.75 ± 0.47	0.60 ± 0.46	NS
Carapace thickness (mm)	0.24 ± 0.08	0.22 ± 0.08	*	0.24 ± 0.08	0.21 ± 0.08	*
PL - right chelae (mm)	19.19 ± 6.27	17.15 ± 5.65	NS	18.55 ± 5.93	16.76 ± 5.78	NS
RC thickness (mm)	0.35 ± 0.10	0.32 ± 0.10	NS	0.34 ± 0.10	0.31 ± 0.10	NS
PL - left chelae (mm)	18.57 ± 6.36	16.65 ± 5.70	NS	17.98 ± 6.07	16.41 ± 5.85	NS
LC thickness (mm)	0.34 ± 0.09	0.31 ± 0.09	NS	0.34 ± 0.09	0.30 ± 0.10	NS
Weight/CW	0.019 ± 0.008	0.016 ± 0.008	NS	0.018 ± 0.008	0.016 ± 0.008	NS
Carapace thickness/CW[a]	5.32 ± 1.08	6.17 ± 1.21	**	5.70 ± 0.84	7.20 ± 1.01	***
RC Thickness/RC[a]	15.56 ± 4.98	17.54 ± 3.77	*	18.85 ± 3.75	19.73 ± 1.92	NS
LC Thickness/LC[a]	16.59 ± 4.01	17.83 ± 3.65	*	17.50 ± 1.91	20.06 ± 1.63	***

[a] Ratio times 1000.

Table 2
Differences in chelae (right and left) ratios (mean SD) of *Carcinus maenas* males and females between immature (CW < 30 mm) and mature (CW > 30 mm) individuals from the lower Minho estuary. S.L., significance level; CW, carapace width; RC, right chelae; LC, left chelae; NS, non-significant; *, S.L. < 0.05; **, S.L. < 0.01; ***, S.L. < 0.001.

Variable	Male			Female		
	Immature	Mature	S.L.	Immature	Mature	S.L.
CW/RC	1.99 ± 0.10	1.90 ± 0.10	**	2.05 ± 0.09	2.03 ± 0.09	NS
CW/LC	2.08 ± 0.07	1.96 ± 0.11	***	2.13 ± 0.10	2.14 ± 0.21	NS
RC thickness/RC[a]	17.91 ± 2.88	16.28 ± 3.64	NS	18.03 ± 2.06	19.17 ± 2.15	*
LC thickness/LC[a]	18.31 ± 2.20	16.38 ± 2.70	**	18.14 ± 1.07	19.59 ± 2.05	**

[a] Times 1000.

cephalothoraxes and chelipeds. Overall, females were proportionally thicker than males, whereas males presented proportionally larger chelae.

Differences in carapace thickness may possibly be related to different life strategies between sexes. Since males might preferentially invest their energy in rapid growth, and hence shedding their exoskeleton faster than females, which preferentially allocate their energy in reproduction, leading them to longer intermoult stages that may ultimately result in a thicker carapace due to the higher accumulation of minerals. An injury to the crab carapace and/or appendages may directly influence moulting, with the regeneration process being concluded after the complete shedding of the damaged carapace (McVean, 1976). Since male crabs are more often engaged in disputes for food and mates than females (Sneddon et al., 2003), they are more susceptible to damages in their carapace and hence they may be forced to moult more often than females. Indeed, the frequency of the ecdysis in males is slightly higher than in females especially when the carapace width is larger than 20 mm (see Crothers, 1967).

Additionally, sexual dimorphism in carapace may be a common trend in crabs (e.g. Lewis, 1977; González-Gurriarán, 1985; Rufino et al., 2004; Ledesma et al., 2010). Usually, males and females differ in carapace shape, with females presenting a larger abdomen that ultimately is beneficial to reproduction (Hartnoll, 1982; Haefner, 2000), while males often present broader carapaces and larger chelae that are advantageous in agonistic encounters (Reid et al., 1994). Thus it is possible that differences in carapace shape

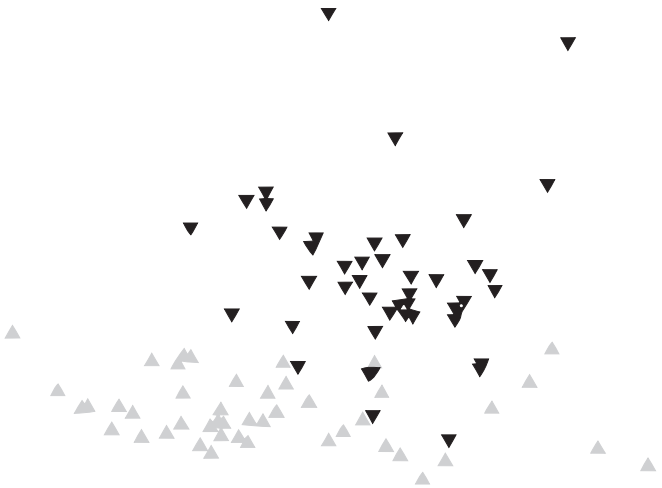


Fig. 4. Two-dimensional nMDS ordination of *Carcinus maenas* carapace (ratios of carapace thickness and weight with width) and chelae variables (ratios of chelae thickness with length) in two sampling sites in Minho estuary. Station 1, grey triangles; station 2, black triangles.

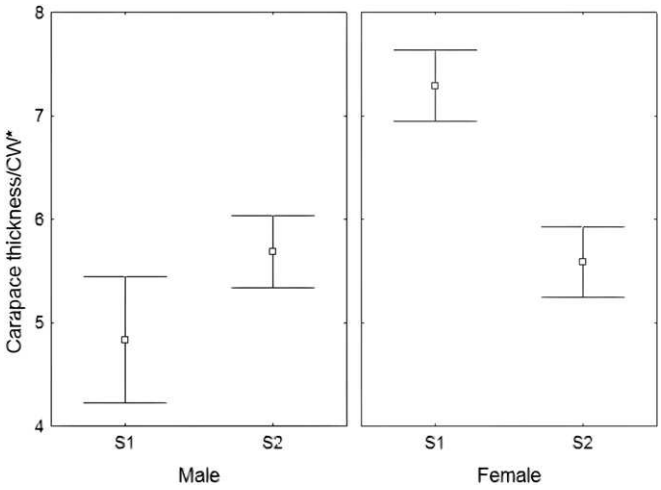


Fig. 5. Ratios (mean SD) of width and thickness of the *Carcinus maenas* carapace at the two sampling stations in lower Minho estuary. CW, carapace width; S1, station 1; S2, station 2.

may also be reflected in carapace structure, with female carapace being relatively thicker than male carapace.

In this study, we also detected significant differences in carapace and cheliped structure between immature and mature individuals. However, there were no differences between males and females during the immature stage. Present observations suggest that, during growth, the carapace of males becomes proportionally thinner, whereas females' becomes thicker. Furthermore, males increase desproportionaly the size of their cheliped, thus, presenting a larger chelae (right and left) when adults, while females did not present differences between the immature and the mature stages. In fact, frequently dominant individuals have larger or stronger chelipeds as shown by other studies (Barki et al., 1997; Sneddon et al., 1997, 2000). Furthermore, allometric growth and sexual dimorphism in cheliped size is common in decapods, with males having larger chelipeds than females (Hartnoll, 1974).

Overall, the red morphotype individuals from all functional groups (i.e. sexes and life stages) and from both sampling stations presented a thicker and heavier carapace, whilst green individuals were lighter and thinner. Several aspects, both physiological and behavioral, differ considerably between color morphotypes including osmoregulation (McGaw and Naylor, 1992), tolerance to hypoxia (Reid and Aldrich, 1989), moulting rate (Reid et al., 1997; Wolf, 1998), growth, mating, muscular and carapace composition and aggressiveness (Reid et al., 1997). Hence it could be expected that green and red morphotypes also differ in carapace structure as was confirmed by our data. Reid et al. (1997) claimed that green and red morphs differ due to a differential use of resources, with green individuals allocating most of their energy in growth, while red

Table 3
Differences in chelae (right and left) ratios (mean SD) of mature (CW > 30 mm) *Carcinus maenas* (males and females) between sampling stations in the lower Minho estuary. S.L., Significance level; CW, carapace width; RC, right chelae; LC, left chelae; NS, non-significant; *, S.L < 0.05; **S.L. < 0.01.

Variable	Male			Female		
	Station 1	Station 2	S.L.	Station 1	Station 2	S.L.
CW/RC	1.86 ± 0.10	1.96 ± 0.08	**	2.04 ± 0.09	2.02 ± 0.08	NS
CW/LC	1.91 ± 0.11	2.02 ± 0.09	**	2.14 ± 0.22	2.11 ± 0.04	NS
RC thickness/RC[a]	16.08 ± 4.29	16.57 ± 2.68	NS	19.35 ± 1.95	17.19 ± 3.67	*
LC thickness/LC[a]	15.50 ± 3.05	17.51 ± 1.69	*	19.83 ± 1.87	16.86 ± 2.36	**

[a] Times 1000.

individuals allocate it to reproduction. Moreover, usually red crabs are dominant over green morphs in aggressive disputes for prey (Keiser et al., 1990). Consequently, the stronger carapace of red crabs reflects its competitive advantage.

4.2. Differences between sites: ecological and behavioral implications

Dissimilarities in carapace variables between sites were observed. In general, females presented a thicker carapace than males, for both color morphtypes. However, in the salt marsh this pattern was somewhat different, with males presenting a slightly thicker carapace than females. It is hard to state the specific reasons that led to this fact although several explanations are possible. The carapace of aquatic crabs is complex in its composition, being constituted by chitin-protein fibers stacked in numerous lamellae that are deeply calcified by minerals such as calcite or amorphous calcium carbonate (Roer and Dillaman, 1984; Giraud-Guille and Bouligand, 1994). Previous studies have acknowledged that the amount of calcium carbonate available in the environment can directly influence the carapace composition of crustaceans (Greenaway, 1983; Cameron, 1985). Also, the water temperature may further influence on the carapace hardening time, being up to 4 times faster in warmer than in colder waters (see Crothers, 1967). Since the water temperature in the salt marsh was higher, it could be expected that the crab carapace probably should be thicker. However, this trend was recorded for males but not for females, suggesting that other factors might be influencing the observed pattern.

The pH value also differed between sampling stations, being higher in the salt marsh. A higher value of the pH might favor the deposition of minerals in the crabs' calcified carapace, since acidic waters usually reduce the calcification rates of calcified marine organisms (Fabry et al., 2008). Thus, it also could be expected that in this sampling station, the carapace of crabs should be thicker than in areas with lower values of pH. On the other hand, since there is more available calcium carbonate in saltier waters (Neufeld and Cameron, 1992), it would be expected that crabs inhabiting areas closer to the sea would present higher amounts of calcium carbonate in its carapace and, thus, a thicker carapace. Therefore, the thicker carapace of females at station 1 could be related to environmental dissimilarities between sampling stations; however, males presented an opposite trend, indicating that other factors might cause males to become thicker at station 2.

After an ecdysis, crustaceans suffer significant calcium loss that needs to be replaced in order to preserve the carapace functions. This calcium uptake is obtained primarily from the environment after moulting, and/or by the use of calcium reservoirs formed before the moult (Greenaway, 1985). The environmental uptake of calcium can occur by the active absorption or transfer from the surrounded water, or by feeding. However, it is believed that feeding may play a less noticeable role in aquatic crabs. Neufeld and Cameron (1992) studying the calcium uptake of the blue crab *Callinectes sapidus*, stated that the diet is not the primary source of calcium carbonate for the species, as the uptake from the surrounded water is the major source of its carapace mineralization. However, they claimed that the mineral re-absorption before the ecdysis and the feeding of exuviae may perhaps play an important role in the calcium carbonate uptake under natural conditions for other crab species.

However, despite the apparently minor importance of the diet in the carapace structure of crabs, its importance cannot be totally disregarded (Neufeld and Cameron, 1992). Differences in food availability at both sampling sites may perhaps further influence the development of a differential carapace structure pattern, because the carapace composition in decapods is related to the individual's diet (Kennish, 1996). Moreover, the diet of *Carcinus maenas* can greatly differ in nearby areas owing mainly to its opportunistic feeding behavior (see Baeta et al., 2006). It may be possible that feeding also contributes to the observed dissimilarities in carapace structure recorded in the present study. On the other hand, *C. maenas* did not present an obvious discrepancy in their diet according to sex and/or size (Baeta et al., 2006), and since males and females presented a distinct trends in their carapace structure, it seems unlike that diet may significantly contributed for the observed dissimilarities recorded. Hence, additional studies are needed in order to elucidate the relative importance of diet in the carapace structure of *C. maenas*.

A well-constituted and thicker carapace will be advantageous in competitive disputes for food or sexual partner. A stronger carapace will provide a better protection, and also, it further enables a stronger and more effective attack tool against conspecifics during feeding and mating interactions. Predator abundance can modulate the expression of defensive morphologies, inducing changes that can reflect within-generation developmental responses (Trussell and Smith, 2000). In this case, it can be expected that when agonistic encounters are more frequent, a stronger carapace will be much more common, and that in crowded areas the agonistic encounters will occur more frequently, contributing to a higher proportion of thicker crabs than is expected.

In the salt marsh, the crab density was twice higher than near the river mouth. Therefore, it is likely that agonistic encounters are more frequent, and thus favouring a thicker carapace, which provides a better protection during disputes. This is especially the case for males, since they usually fight over resources (food, individual space, females), with the larger males being dominant (Berrill and Arsenault, 1982). Fights between males are significantly more intense in the presence of females (Sneddon et al., 2003) than of food. These fights are costly with 40% of fighting crabs incurring an injury (Fletcher and Hardege, 2009). Therefore in a highly densely occupied space, such as in the Coura salt marsh, it is expected that males with a thicker and stronger carapace will have more easily access to food and also to females. Intraspecific competition could then be leading to thicker males as a pattern.

In decapod crustaceans, the first pereiopod generally bears a chelae (claw) formed by a modification of the dactylus and propodus which is known as a cheliped. This appendage is typically used in feeding, as well as a defense tool against predators and during agonistic behavior, and hence has also been considered a multifunctional organ (Brown et al., 1979; Lee, 1995) which can also be highly variable. Mature males presented significantly larger cheliped than immature males, whereas this was not registered in females. Males further presented a larger ratio of CW with chelae size (right and left) near the river mouth, while females did not differ. These results can be due to intraspecific competition for food in the case of immature individuals. In respect to mature males, the stronger chelae might be a determinant in the competition for mating, since during agonistic encounters males with larger chelipeds often become dominant over smaller individuals (Sneddon et al., 1997, 2000). In fact, in the Minho estuary, ovigerous females are exclusively recorded in areas near the river mouth of this estuary (ATS, personal observation).

This study further highlighted that even in nearby areas within an ecosystem, significant morphological differences occurs. These differences probably were related to ecological and behavioral dissimilarities among microhabitats, thus it is important to conduct further researches that address a wide ecological and behavioral spectrum, aiming to assess the importance of each feature in the ecomorphology of the species.

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